

$$I) \frac{Y}{L} = \frac{A K^{\alpha} L^{1-\alpha}}{L} = A \left(\frac{K}{L}\right)^{\alpha} \Rightarrow Y = A K^{\alpha}$$

II) NORTE:

$$K_N = K_{N0}/L_{N0} = 100/10 = 10 \checkmark$$

$$Y_{N0} = A_N K_N^{1/4} = 4 \cdot 10^{1/4} = 7,1131 \checkmark \quad Y_N = 71,13$$

$$IV) SUL: K_S = K_{S0}/L_{S0} = \frac{4000}{20} = 200 \checkmark$$

$$Y_{S0} = A_S \cdot K_S^{1/2} = 5 \cdot 200^{1/2} = 100 \checkmark \quad Y_S = 2000$$

~~$$I) K_{t+1} = K_T = K^*$$~~

$$I) K^* = \frac{(1-\delta)K^* + S A K^{*\alpha}}{1+n}$$

$$(1+\delta) K^* = S A K^{*\alpha}$$

$$II) K^* = \left[\frac{S A}{1+\delta} \right]^{1/(1-\alpha)} \checkmark$$

III) SUBST.

$$Y^* = A K^{*\alpha}$$

$$Y^* = A \left[\frac{S A}{1+\delta} \right]^{\frac{\alpha}{1-\alpha}}$$

IV) SUBST. VALUES

$$B = \frac{S_N A_N}{n_N + \delta} = \frac{0,16 \cdot 4}{0,03 + 0,05} = \frac{0,64}{0,08} = 8; \quad \frac{\alpha}{1-\alpha_N} = \frac{1}{0,75} = \frac{4}{3}$$

$$K_N^* = B^{\frac{1}{\alpha}} = 8^{1/3} = 2 \checkmark; \quad Y_N^* = 4 \cdot K_N^{1/4} = 4 \cdot 2 = 8 \checkmark$$

$$D) \beta_s = \frac{s A_s}{m_s + \delta} = \frac{0,3 \cdot 5}{0,11 + 0,05} = \frac{1,5}{0,06} = 25 \quad \text{e} \quad \sigma_s = \frac{1}{1-\alpha} = \frac{1}{1-0,6} = 2$$

$$K_s^* = \beta_s^{\sigma_s} = 25^2 = 625 \quad \checkmark$$

$$Y_s^* = A_s \cdot K_s^{\alpha_s} = 5 \cdot 625^{\frac{1}{2}} = 5 \cdot 25 = 125 \quad \checkmark$$

O que se ver, é uma divergência dos níveis de output das economias ~~X~~

~~C)~~ UNIFICANDO

$$K = 800 + 100 = 900 \quad ; \quad L = 20 + 10 = 30 \quad ; \quad k = \frac{900}{30} = 30$$

$$I) Y = A_s K^{\alpha_s} = 5 \cdot (900)^{\frac{1}{2}} = 82,1584 \quad \checkmark, \quad Y = 82,1584 \times 30 = 2464,75$$

II) O SUL PERDE GDP per CAPITA ~~X~~ $\checkmark$

A SEMA DO AGREGADO AUMENTA

$$Y_{S_0} + Y_{N_1} = 2000 + 71,13 = 2071,13 \quad \checkmark$$

$$\del{C) I) m} = \frac{L_N m_N + L_S m_S}{L_N + L_S} = \frac{10 \cdot 0,03 + 20 \cdot 0,01}{10 + 20} = \frac{0,5}{30} = \frac{1}{60}$$

$$II) \beta = \frac{s \cdot A_s}{m + \delta} = \frac{0,3 \cdot 5}{\frac{1}{60} + \frac{1}{20}} = \frac{0,3 \cdot 5 \cdot 60}{4} = 22,5$$

$$\text{III)} \quad K^* = B \frac{1}{1-\alpha} = 22,5^2 = 506,25$$

$$Y^* = A \cdot K^{*\alpha} = 5 \cdot (22,5^2)^{\frac{1}{2}} = 112,5$$

IV) SIM CLA ESTA ABAIXO DO ESTADO ESTACIONÁRIO

$$\frac{y_u}{y_s} = \frac{112,5}{125} = 0,90$$

$$\frac{K_u}{K_s} = \frac{506,25}{625} = 0,81$$

$$2) \checkmark \bar{K} = \bar{K}_p + \theta \bar{K}_p = (1+\theta) \bar{K}_p$$

$$ii) \bar{K}_p = \frac{\bar{K}}{1+\theta}$$

$$iii) Y = A \bar{K}_p^\alpha L^{1-\alpha} = A (1+\theta)^{-\alpha} \bar{K}^\alpha L^{1-\alpha} \checkmark$$

$$\bar{K}_s = \theta \bar{K}_p = \frac{\theta}{1+\theta} \bar{K}$$

$$\left(\frac{\bar{K}_s}{\bar{K}_p} = \frac{\theta}{1+\theta} \right) \checkmark$$

iv) ANALISANDO PCADA:

$$\theta = 0.16$$

$$\frac{Y(\theta)}{Y(0)} = \frac{A (1+\theta)^{-\alpha} \bar{K}^\alpha L^{1-\alpha}}{A \bar{K}^\alpha L^{1-\alpha}} = (1+\theta)^{-\alpha} = 1.16^{-1/3} = 0.8736$$

PCADA DE OUTPUT.

$$(0.1264) \checkmark$$

$$\checkmark i) \pi = A \bar{K}_p^\alpha L^{1-\alpha} - \underbrace{\eta(1+\theta) \bar{K}_p}_{\bar{K}} - wL$$

$$ii) \frac{\partial \pi}{\partial \bar{K}_p} = \alpha \underbrace{A \bar{K}_p^{\alpha-1} L^{1-\alpha}}_{\alpha \frac{Y}{\bar{K}_p}} - \eta(1+\theta) = 0$$

$$\alpha \frac{Y}{\bar{K}_p} = \eta(1+\theta)$$

$$\eta = \alpha \frac{Y}{\bar{K}_p(1+\theta)} = \alpha \frac{Y}{\bar{K}} \quad (\text{NÃO DISTORCE})$$

$$iii) \frac{\partial \pi}{\partial L} = (1-\alpha) A \bar{K}_p^\alpha L^{-\alpha} - w = 0$$

$$(1-\alpha) A \frac{Y}{L} - w = 0$$

$$w = (1-\alpha) A \frac{Y}{L} \checkmark$$

$$i) \quad f_K = \frac{rK}{Y} = \frac{\alpha \frac{Y}{K} \cdot K}{Y} = \alpha$$

$$f_L = \frac{wL}{Y} = \frac{(1-\alpha) \frac{Y}{L} \cdot L}{Y} = 1-\alpha$$

$$ii) \quad i) \quad Y = \hat{A} K^\alpha L^{1-\alpha} \Rightarrow \hat{A} = \frac{Y}{K^\alpha L^{1-\alpha}}$$

$$ii) \quad \hat{A} = \frac{A(1+\theta)^{-\alpha} K^\alpha L^{1-\alpha}}{K^\alpha L^{1-\alpha}} = A(1+\theta)^{-\alpha} \checkmark$$

$$iii) \quad \frac{\hat{A}}{A} = (1+\theta)^{-\alpha} = (1,5)^{-\frac{1}{3}} = 0,8736 \checkmark$$

d) X